



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-042 Protected Practice Review

Open only the matching Rxx after a saved Pxx attempt. Then close review and complete changed retry.

Artifact	C3-042
Status	2026 - LEARNER EDITION
Phase	C3-03 Orchestration & Reliable Workflows 13 lessons 34-46 focused hours
Prerequisite	C3-02 Gate PASS; C2B Python/quality/ownership foundations assumed, orchestration taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: simple mental picture -> tiny example -> guided Airflow/tool workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A green scheduler box alone never proves correct data.



Protection rule

This file is not normal reading. Review reveals repair guidance after effort. It must never be used to create the first attempt. Original attempts remain in history.

R01 - Workflow orchestration and DAG design

OPEN ONLY AFTER P01 IS SAVED.

What strong evidence looks like

Every task has a single operational responsibility; publish is blocked by validation/reconciliation; no task depends on hidden local leftovers.

Baseline repair guidance

R01 - Review for C3-03-L01 Expected reasoning shape A useful DAG boundary has an independently observable outcome and a meaningful retry/recovery boundary. Evidence that should exist If validation fails, publish must not run; rerunning validation/publish should not require re-ingesting correct source. Common weak patterns Green scheduler/UI without data/state validation; wall-clock date substituted for logical interval; hidden global state; retries used to mask deterministic defects; duplicate side effects; secrets in source/evidence; full-DAG rerun chosen without checking partial state. Repair route Return only to the failed competency. Preserve the failed evidence, change one dimension of the problem, and rerun a micro-check. Do not restart C3-03 unless multiple independent foundations are missing. Changed fresh check Change the upstream source type and redesign only the boundaries that truly change. PASS standard Correct + reproducible + interval-aware + idempotent where side effects exist + independently validated + explainable without this review open.

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Changed retry

Redesign the same workflow when extraction comes from an API but loading still targets the same warehouse. Defend which task boundaries change and which do not.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and the saved evidence can be explained. A narrow weakness routes to targeted repair; it never auto-unlocks a fresh full Gate retest.

R02 - Airflow setup, standalone mode and project structure

OPEN ONLY AFTER P02 IS SAVED.

What strong evidence looks like

Airflow reports 3.3.1; standalone starts; source project is separate from metadata/logs/secrets; no credential appears in Git-tracked files.

Baseline repair guidance

R02 - Review for C3-03-L02 Expected reasoning shape Local standalone proves a development environment, not production topology. Durable source/config expectations must survive deleted runtime state. Evidence that should exist Recreate venv/runtime and rediscover the DAG without restoring metadata DB/logs/credentials. Common weak patterns Green scheduler/UI without data/state validation; wall-clock date substituted for logical interval; hidden global state; retries used to mask deterministic defects; duplicate side effects; secrets in source/evidence; full-DAG rerun chosen without checking partial state. Repair route Return only to the failed competency. Preserve the failed evidence, change one dimension of the problem, and rerun a micro-check. Do not restart C3-03 unless multiple independent foundations are missing. Changed fresh check Recreate in a fresh project path and compare source/runtime separation. PASS standard Correct + reproducible + interval-aware + idempotent where side effects exist + independently validated + explainable without this review open.

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Changed retry

Delete only the disposable venv/runtime state, recreate the environment, and prove the project source still parses/discovers.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and the saved evidence can be explained. A narrow weakness routes to targeted repair; it never auto-unlocks a fresh full Gate retest.



R03 - DAG authoring with TaskFlow and tasks

OPEN ONLY AFTER P03 IS SAVED.

What strong evidence looks like

DAG imports from `airflow.sdk`; import time has no I/O side effects; business logic remains testable outside Airflow; dependencies match the intended graph.

Baseline repair guidance

R03 - Review for C3-03-L03 Expected reasoning shape Airflow 3 DAG authoring should use public `airflow.sdk`; pure logic belongs outside the DAG; XCom carries compact metadata. Evidence that should exist Unit tests for `src/` can run without scheduler; DAG import has no side-effect I/O. Common weak patterns Green scheduler/UI without data/state validation; wall-clock date substituted for logical interval; hidden global state; retries used to mask deterministic defects; duplicate side effects; secrets in source/evidence; full-DAG rerun chosen without checking partial state. Repair route Return only to the failed competency. Preserve the failed evidence, change one dimension of the problem, and rerun a micro-check. Do not restart C3-03 unless multiple independent foundations are missing. Changed fresh check Add one metadata output and preserve the no-bulk-XCom rule. PASS standard Correct + reproducible + interval-aware + idempotent where side effects exist + independently validated + explainable without this review open. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-03 PROTECTED REVIEW

Changed retry

Add a second compact metadata output (for example `row_count`) without passing the underlying dataset through XCom.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and the saved evidence can be explained. A narrow weakness routes to targeted repair; it never auto-unlocks a fresh full Gate retest.



R04 - Schedules, logical dates and data intervals

OPEN ONLY AFTER P04 IS SAVED.

What strong evidence looks like

Historical run always maps to historical input/output; two intervals cannot overwrite each other; start date is timezone-aware; catchup choice is documented.

Baseline repair guidance

R04 - Review for C3-03-L04 Expected reasoning shape Business partition comes from the logical/data interval, not wall-clock today. Evidence that should exist Rerunning an old logical date resolves the same historical input/output partition. Common weak patterns Green scheduler/UI without data/state validation; wall-clock date substituted for logical interval; hidden global state; retries used to mask deterministic defects; duplicate side effects; secrets in source/evidence; full-DAG rerun chosen without checking partial state. Repair route Return only to the failed competency. Preserve the failed evidence, change one dimension of the problem, and rerun a micro-check. Do not restart C3-03 unless multiple independent foundations are missing. Changed fresh check Change schedule/grain and restate one-run meaning. PASS standard Correct + reproducible + interval-aware + idempotent where side effects exist + independently validated + explainable without this review open.

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Changed retry

Change the schedule from daily to a different interval and restate what one run represents before editing path logic.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and the saved evidence can be explained. A narrow weakness routes to targeted repair; it never auto-unlocks a fresh full Gate retest.



R05 - Parameters, variables, connections and secret boundaries

OPEN ONLY AFTER P05 IS SAVED.

What strong evidence looks like

No secret literal in tracked source; per-run values are explicit; connection identifiers are stable; config ownership/precedence is documented.

Baseline repair guidance

R05 - Review for C3-03-L05 Expected reasoning shape Params = run input; Variables/env = deployment/runtime config; Connections = external-system config; credentials stay out of source. Evidence that should exist Repository scan has no secret literal and code moves environments via external config. Common weak patterns Green scheduler/UI without data/state validation; wall-clock date substituted for logical interval; hidden global state; retries used to mask deterministic defects; duplicate side effects; secrets in source/evidence; full-DAG rerun chosen without checking partial state. Repair route Return only to the failed competency. Preserve the failed evidence, change one dimension of the problem, and rerun a micro-check. Do not restart C3-03 unless multiple independent foundations are missing. Changed fresh check Move local -> second environment with no DAG code edit. PASS standard Correct + reproducible + interval-aware + idempotent where side effects exist + independently validated + explainable without this review open. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-03 PROTECTED REVIEW

Changed retry

Move the same DAG from local to another environment without changing DAG code; list which environment-owned values must be supplied.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and the saved evidence can be explained. A narrow weakness routes to targeted repair; it never auto-unlocks a fresh full Gate retest.

R06 - Retries, timeouts and failure callbacks

OPEN ONLY AFTER P06 IS SAVED.

What strong evidence looks like

Transient path shows failed attempt then success; permanent bad-data path stays failed; side effects remain idempotent; failure log retains run/task/partition context.

Baseline repair guidance

R06 - Review for C3-03-L06 Expected reasoning shape Retry only a failure that may become successful on a later attempt without data/code correction; permanent bad input fails clearly. Evidence that should exist Fail-once path shows failure then success; malformed timestamp remains failed; no duplicate side effect. Common weak patterns Green scheduler/UI without data/state validation; wall-clock date substituted for logical interval; hidden global state; retries used to mask deterministic defects; duplicate side effects; secrets in source/evidence; full-DAG rerun chosen without checking partial state. Repair route Return only to the failed competency. Preserve the failed evidence, change one dimension of the problem, and rerun a micro-check. Do not restart C3-03 unless multiple independent foundations are missing. Changed fresh check Swap dependency to rate-limited API and redesign retry/backoff. PASS standard Correct + reproducible + interval-aware + idempotent where side effects exist + independently validated + explainable without this review open. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-03 PROTECTED REVIEW Page 8

Changed retry

Change the dependency from local file to rate-limited API and redesign retry/backoff without changing the business validation policy.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and the saved evidence can be explained. A narrow weakness routes to targeted repair; it never auto-unlocks a fresh full Gate retest.



R07 - Catchup, backfills and controlled reprocessing

OPEN ONLY AFTER P07 IS SAVED.

What strong evidence looks like

Correct historical partitions; no duplicate current keys; newest valid version retained; rerun materially unchanged; final totals reconcile.

Baseline repair guidance

R07 - Review for C3-03-L07 Expected reasoning shape Backfill is deliberate history recreation with explicit date range/reprocess behavior/concurrency; output must be partition-safe and idempotent. Evidence that should exist O7002 retains the newer version after older reprocessing; rerun does not duplicate keys. Common weak patterns Green scheduler/UI without data/state validation; wall-clock date substituted for logical interval; hidden global state; retries used to mask deterministic defects; duplicate side effects; secrets in source/evidence; full-DAG rerun chosen without checking partial state. Repair route Return only to the failed competency. Preserve the failed evidence, change one dimension of the problem, and rerun a micro-check. Do not restart C3-03 unless multiple independent foundations are missing. Changed fresh check Use failed-only reprocessing and explain the difference. PASS standard Correct + reproducible + interval-aware + idempotent where side effects exist + independently validated + explainable without this review open. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-03 PROTECTED REVIEW

Changed retry

Repeat with `failed`-only reprocessing and explain why it differs from deliberately recreating completed runs.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and the saved evidence can be explained. A narrow weakness routes to targeted repair; it never auto-unlocks a fresh full Gate retest.

R08 - Sensors and data-aware dependencies

OPEN ONLY AFTER P08 IS SAVED.

What strong evidence looks like

Downstream is blocked before readiness; present input proceeds; missing input fails within bound; waiting does not use a custom busy loop.

Baseline repair guidance

R08 - Review for C3-03-L08 Expected reasoning shape Readiness is an explicit condition with timeout and efficient wait behavior. Evidence that should exist Present input releases downstream; missing input times out/fails; no busy loop. Common weak patterns Green scheduler/UI without data/state validation; wall-clock date substituted for logical interval; hidden global state; retries used to mask deterministic defects; duplicate side effects; secrets in source/evidence; full-DAG rerun chosen without checking partial state. Repair route Return only to the failed competency. Preserve the failed evidence, change one dimension of the problem, and rerun a micro-check. Do not restart C3-03 unless multiple independent foundations are missing. Changed fresh check Replace file readiness with an upstream marker/table readiness rule. PASS standard Correct + reproducible + interval-aware + idempotent where side effects exist + independently validated + explainable without this review open. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-03 PROTECTED REVIEW

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Changed retry

Replace filesystem readiness with an upstream marker/table condition and preserve the same readiness/timeout contract.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and the saved evidence can be explained. A narrow weakness routes to targeted repair; it never auto-unlocks a fresh full Gate retest.



R09 - Idempotent task design and partial reruns

OPEN ONLY AFTER P09 IS SAVED.

What strong evidence looks like

Same logical input twice gives same business state; no duplicate keys; current version rule preserved; partial rerun touches only intended outputs.

Baseline repair guidance

R09 - Review for C3-03-L09 Expected reasoning shape Same logical input must converge to same side-effect state; stable keys/version rules and atomic publish matter. Evidence that should exist Run twice -> same keys/current versions/summary; partial rerun does not double count. Common weak patterns Green scheduler/UI without data/state validation; wall-clock date substituted for logical interval; hidden global state; retries used to mask deterministic defects; duplicate side effects; secrets in source/evidence; full-DAG rerun chosen without checking partial state. Repair route Return only to the failed competency. Preserve the failed evidence, change one dimension of the problem, and rerun a micro-check. Do not restart C3-03 unless multiple independent foundations are missing. Changed fresh check Inject a stale version and prove current state does not regress. PASS standard Correct + reproducible + interval-aware + idempotent where side effects exist + independently validated + explainable without this review open. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-03 PROTECTED REVIEW

Changed retry

Change one source row to a stale earlier version and prove replay does not regress the current record.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and the saved evidence can be explained. A narrow weakness routes to targeted repair; it never auto-unlocks a fresh full Gate retest.

R10 - Pools, concurrency and resource limits

OPEN ONLY AFTER P10 IS SAVED.

What strong evidence looks like

Tasks consuming the scarce resource share an explicit pool; limits are justified; a backfill cannot launch unbounded concurrent writes.

Baseline repair guidance

R10 - Review for C3-03-L10 Expected reasoning shape Pools/concurrency protect a named scarce resource; limits must be justified by dependency capacity. Evidence that should exist Database-writing tasks share pool; backfill active runs are bounded. Common weak patterns Green scheduler/UI without data/state validation; wall-clock date substituted for logical interval; hidden global state; retries used to mask deterministic defects; duplicate side effects; secrets in source/evidence; full-DAG rerun chosen without checking partial state. Repair route Return only to the failed competency. Preserve the failed evidence, change one dimension of the problem, and rerun a micro-check. Do not restart C3-03 unless multiple independent foundations are missing. Changed fresh check Assume API capacity halves and redesign limits. PASS standard Correct + reproducible + interval-aware + idempotent where side effects exist + independently validated + explainable without this review open. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-03 PROTECTED REVIEW

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Changed retry

Move the same workflow to a slower API and redesign the pool/concurrency limits from the dependency's contract.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and the saved evidence can be explained. A narrow weakness routes to targeted repair; it never auto-unlocks a fresh full Gate retest.

R11 - DAG testing and pre-deployment validation

OPEN ONLY AFTER P11 IS SAVED.

What strong evidence looks like

Pure tests pass; DAG compiles/imports; no import-time external work; dependency graph is intentional; runtime checks are clearly distinguished from static checks.

Baseline repair guidance

R11 - Review for C3-03-L11 Expected reasoning shape Layered tests prove different things: compile/static, pure unit logic, DAG import/graph and actual Airflow runtime. Evidence that should exist No import-time I/O; tests pass; DAG discovery/test succeeds on learner runtime. Common weak patterns Green scheduler/UI without data/state validation; wall-clock date substituted for logical interval; hidden global state; retries used to mask deterministic defects; duplicate side effects; secrets in source/evidence; full-DAG rerun chosen without checking partial state. Repair route Return only to the failed competency. Preserve the failed evidence, change one dimension of the problem, and rerun a micro-check. Do not restart C3-03 unless multiple independent foundations are missing. Changed fresh check Introduce a new failing pure-function fixture and show CI/predeploy blocks it. PASS standard Correct + reproducible + interval-aware + idempotent where side effects exist + independently validated + explainable without this review open. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-03 PROTECTED REVIEW Page 13

Changed retry

Add a fresh failing fixture to a pure function test and show that pre-deploy QA blocks the change before the DAG is scheduled.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and the saved evidence can be explained. A narrow weakness routes to targeted repair; it never auto-unlocks a fresh full Gate retest.



R12 - Operational logs, metadata and troubleshooting

OPEN ONLY AFTER P12 IS SAVED.

What strong evidence looks like

Diagnosis names the failure layer and evidence; recovery is minimal and idempotent; no secret values are copied into evidence.

Baseline repair guidance

R12 - Review for C3-03-L12 Expected reasoning shape Troubleshooting must name run/task/interval/state and classify failure layer before recovery. Evidence that should exist Diagnosis distinguishes parse, wait, transient runtime and permanent data defects; recovery is minimal. Common weak patterns Green scheduler/UI without data/state validation; wall-clock date substituted for logical interval; hidden global state; retries used to mask deterministic defects; duplicate side effects; secrets in source/evidence; full-DAG rerun chosen without checking partial state. Repair route Return only to the failed competency. Preserve the failed evidence, change one dimension of the problem, and rerun a micro-check. Do not restart C3-03 unless multiple independent foundations are missing. Changed fresh check Diagnose queued-not-running caused by pool saturation. PASS standard Correct + reproducible + interval-aware + idempotent where side effects exist + independently validated + explainable without this review open. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-03 PROTECTED REVIEW
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Changed retry

Diagnose a fresh 'queued but not running' case caused by a deliberately constrained pool rather than code failure.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and the saved evidence can be explained. A narrow weakness routes to targeted repair; it never auto-unlocks a fresh full Gate retest.



R13 - Deployment patterns, recovery runbooks and handoff

OPEN ONLY AFTER P13 IS SAVED.

What strong evidence looks like

No secrets/runtime state bundled; version/source identity recorded; deployment and recovery are reproducible; runbook includes validation after recovery.

Baseline repair guidance

R13 - Review for C3-03-L13 Expected reasoning shape Handoff includes deployment identity, config, verification, incident detection, recovery/reprocess, validation and ownership. Evidence that should exist Second person can recover two known incidents without author assistance and prove state afterward. Common weak patterns Green scheduler/UI without data/state validation; wall-clock date substituted for logical interval; hidden global state; retries used to mask deterministic defects; duplicate side effects; secrets in source/evidence; full-DAG rerun chosen without checking partial state. Repair route Return only to the failed competency. Preserve the failed evidence, change one dimension of the problem, and rerun a micro-check. Do not restart C3-03 unless multiple independent foundations are missing. Changed fresh check Add a task-ID/config migration and write rollback/migration notes. PASS standard Correct + reproducible + interval-aware + idempotent where side effects exist + independently validated + explainable without this review open.

Changed retry

Assume the next release changes one task ID and one connection requirement. Write the migration/rollback note that prevents accidental historical confusion.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and the saved evidence can be explained. A narrow weakness routes to targeted repair; it never auto-unlocks a fresh full Gate retest.